

especially since the list of device drivers kept increasing. Also, over 18,000 device files consumed too much extra space on the disk. These problems were resolved by adding the ability to ignore the major and minor numbers in `udev`.

Particularly for hot-pluggable devices like USB hardware, there was no consistency in the naming/mapping of device files to the actual hardware device. For instance, on a system with two USB printers, one might be called `/dev/usb/lp0` and the other `/dev/usb/lp1`—but it was never certain which printer was referred to by which device node. That could change based on which one was powered on when the computer booted, or was connected first—or could change if they were connected to a USB hub instead of directly to the system USB ports. This often proved frustrating and confusing to users. `udev` provides a means to assign a persistent name to a device, using the `udev` rules feature.

Other `udev` features resolve many problems inherent to `devfs`:

- `udev` works in user-space, reducing kernel code and complexity.
- `udev` provides a way to assign persistent names to devices, independent of the order of connection or location on a bus.
- `udev` dynamically populates the `/dev` directory with device nodes for only the devices that are actually present/connected to the system. It also enables the assignment of common names to devices via symbolic links to the actual device node.
- `udev` provides complete device information to user-space

programs, removing the need to access kernel-space for the information.

How `udev` works

The `udev` daemon listens to a `netlink` socket for `uevents` issued by the kernel on the connection or removal of a device. You can monitor these events with the command `udevmonitor`—run it, attach a USB device like a flash drive, and remove it. (In newer distributions, `udevmonitor` may not be available—in that case, use `udevadm` monitor instead of `udevmonitor`.)

At start-up, `udev` mounts a `tmpfs` filesystem for the `/dev` directory. It then copies device nodes from `/lib/udev/device` to `/dev` and begins collecting `uevents` from the kernel for cold-plugged devices. `/etc/udev/rules.d` is used to apply device parameters, to create symbolic links, and to take other action. For hot-plugged devices, `udev` catches the `uevent` via D-Bus, and matches the attributes of the new device as exposed in the `/sys` directory, with `udev` rules—and then creates the device node in the `/dev` directory. `udev` can also load properly-written device drivers for detected hardware, using the `modalias` mechanism.

`udev` rules and examples

`udev` gives you the power to handle your devices by writing your own rules and configuration files. You can override the behaviour of package-supplied rules (usually found in `/lib/udev/rules.d`), or add custom and specific behaviour to suit your

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